

DEP SPECIFICATION

ONSHORE CONCRETE DESIGN AND CONSTRUCTION

DEP 34.19.20.31-Gen.

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DESIGN AND ENGINEERING PRACTICE



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1. INTRODUCTION

1.1 SCOPE

This DEP specifies requirements and gives recommendations for the design and execution of onshore concrete structures. This DEP also includes the requirements and recommendations for chemical-resistant protection systems for concrete surfaces.

This DEP applies to both in-situ work and construction using manufactured/prefabricated concrete elements.

DEPs are intended for use in oil and gas exploration, production and transport, oil refining, chemical processing, supply and distribution. This DEP also can be applied in other similar facilities including non-hydrocarbon energy applications.

This is a revision of the DEP of the same number dated February 2025; see (1.5) regarding the changes.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by Shell GSI, the distribution of this DEP is confined to Shell companies and, where necessary, to Contractors and Manufacturers/Suppliers nominated by them. Any authorised access to DEPs does not for that reason constitute an authorisation to any documents, data or information to which the DEPs refer.

When DEPs are applied, a Management of Change (MOC) process is necessary. MOC is of particular importance when modifying existing facilities.

If national or local regulations exist in which some of the requirements could be more stringent than in this DEP, the Contractor shall determine which of the requirements are the more stringent and which combination of requirements will be acceptable with regards to the safety, environmental, economic and legal aspects. In all cases, the Contractor shall inform the Principal of any deviation from the requirements of this DEP which is considered to be necessary in order to comply with national or local regulations. The Principal can then negotiate with the Authorities concerned, the objective being to obtain agreement to follow this DEP as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Contractor** is the party that carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party that manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The **Principal** is the party that initiates the project and ultimately pays for it. The Principal may also include an agent or consultant authorised to act for, and on behalf of, the Principal.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

The word **may** indicates a permitted option.

1.3.2 Specific definitions

The following definitions are specific to this DEP.

Term	Definition
Applicator	Construction Contractor responsible for application of Chemical resistant protective systems
Combined System	Typically, these systems are called acid-proof brick or chemical resistant masonry unit systems or brick lining systems. Such systems comprise a sealing layer and a service layer. Taken together, the two layers provide a more effective protection than each layer would provide on its own.
Liquids harmful to concrete	Substances that can chemically or physically degrade the integrity, strength, or durability of concrete.
Mass Concreting	Concrete pours subject to limited external restraining, such as soil bearing foundations cast on blinding and pile caps with thickness more than 1200 mm (48 in) and concrete members subject to high external restraining, such as concrete walls with thickness more than 500 mm (20 in) poured on hardened concrete slabs.
Pot Life	Time during which the coating material can be readily used.

1.3.3 Abbreviations

The following abbreviations are specific to this DEP.

Term	Definition
ITP	Inspection and Test Plan
LNG	Liquefied Natural Gas
NDT	Non-Destructive Testing (also known as NDE)
QCR	Quality Control Records
SCC	Self-Compacting Concrete
SSPC	The Society for Protective Coatings
UPV	Ultrasonic Pulse Velocity
WCR	Water Cement Ratio

1.4 CROSS-REFERENCES

Where cross-references to other parts of this DEP are made, the referenced section or clause number is shown in brackets (). Other documents referenced by this DEP are listed in (7).

1.5 SUMMARY OF MAIN CHANGES

This DEP is a revision of the DEP of the same number dated February 2025. The following are the main non-editorial changes.

Section/Clause	Change
1	Updated the scope statement to establish that this DEP also includes the requirements and recommendations for chemical resistant protection systems.
2.9	New section that provides content for design requirements for chemical resistant protective systems that previously was in Section 2 and Section 3.1 of DEP 30.48.60.12-Gen. (which has been withdrawn for the February 2026 publication of DEPs).
3.7	New section covering chemical resistant combined lining systems that previously was in Section 3.2 of DEP 30.48.60.12-Gen. (which has been withdrawn for the February 2026 publication of DEPs).
4.16	New section that provides content for chemical resistant protection systems that previously was in Section 4 of DEP 30.48.60.12-Gen. (which has been withdrawn for the February 2026 publication of DEPs).
5	New section that provides content for chemical resistant protection systems that previously was in Section 5 of DEP 30.48.60.12-Gen. (which has been withdrawn for the February 2026 publication of DEPs).
6.3	New section that provides content for chemical resistant protection systems that previously was in Section 6 of DEP 30.48.60.12-Gen. (which has been withdrawn for the February 2026 publication of DEPs).

1.6 COMMENTS ON THIS DEP

Comments on this DEP may be submitted to the Administrator using one of the following options:

Shell DEPs Online (Users with access to Shell DEPs Online)	Enter the Shell DEPs Online system at https://www.shelldeps.com Select a DEP and then go to the details screen for that DEP. Click on the "Give feedback" link, fill in the online form and submit.
DEP Feedback System (Users with access to Shell Network)	Enter comments directly in the DEP Feedback System which is accessible from the Technical Standards Portal. Select "Submit DEP Feedback", fill in the online form and submit.
Offline Feedback Form (other users)	Use Standard Form DEP 00.00.05.80-Gen. to record feedback and email the form to the Administrator at standards@shell.com .

Feedback that has been registered in the DEP Feedback System by using one of the above options will be reviewed by the DEP Custodian for potential improvements to the DEP.

1.7 DUAL UNITS

This DEP contains both the International System (SI) units, as well as the corresponding US Customary (USC) units, which are given following the SI units in brackets. When agreed by the Principal, the indicated USC values/units may be used.

1.8 NON NORMATIVE TEXT (COMMENTARY)

Text shown in italic style in this DEP indicates text that is non-normative and is provided as explanation or background information only.

Non-normative text is normally indented slightly to the right of the relevant DEP clause.

2. DESIGN OF CONCRETE STRUCTURES

2.1 LOADS AND LOAD COMBINATIONS

1. The loads and load combinations for onshore concrete structures shall be in accordance with one of the following, whichever applies:
 - a. DEP 35.00.10.10-Gen. for nearshore and coastal concrete structures;
 - b. DEP 34.00.01.30-Gen. for onshore concrete structures other than nearshore and coastal concrete structures.

Rationale: To ensure consistency across DEPs.

2.2 CONSTRUCTION AND MOVEMENT JOINTS

1. Construction and movement joints shall be clearly specified and indicated in the concrete design drawings.

Rationale: For the integrity of the structure, it is important that all construction and movement joints are defined in construction documents and constructed as required.

2.3 ANCHOR BOLTS AND EMBEDMENTS

1. Anchor bolts design and construction shall be in accordance with applicable local codes.
 - a. In the absence of local codes, ASCE guidelines shall be followed as a minimum.
2. For foundations supporting static equipment, the Manufacturer/Supplier shall provide one of the following:
 - a. Preferred Option: an anchor bolt template;
 - b. Subject to Principal approval: concrete pockets for situations where an anchor bolt template is not provided, to allow for minor (i.e., less than 6 mm (0.25 in)) infield adjustment.

Rationale: Each equipment or structure has an accurate bolt pattern in the bottom plates, which anchor bolts have to meet in installation phase. Any mismatch of anchor bolts with baseplate holes could have a major design, cost and schedule impact. To ensure accurate fit, use of a template that has the same bolt pattern as the equipment bottom plates is recommended. Concrete pockets require more space, and therefore their use requires Principal approval.

3. For foundations supporting vibrating equipment, anchor bolts shall be installed using sleeves, unless otherwise specified by the Principal.

Rationale: Sleeves are provided for vibrating equipment when anchor bolts have to be pre-tensioned in order to maintain the bolts under continuous tensile stresses during potential load reversals from vibrations of equipment.

4. Anchor bolts subject to uplift or vibration shall be locked with a washer and two nuts.

Rationale: An additional nut on a bolt or stud joint is used to prevent the main nut from loosening due to vibrations and it ensures adequate transfer of uplift/tension forces.

5. Anchor bolts and assemblies shall comply with Standard Drawing S 10.047.

Rationale: Following this drawing will ensure proper installation of the anchor bolts.

6. Post-installed anchor bolts shall be hot dip galvanised according to DEP 34.28.00.31-Gen.

Rationale: Hot dip galvanisation enhances the bolt's corrosion resistance and hence increases its longevity.

7. Aluminium items shall not be used for concrete embedment.

Rationale: Aluminium can react with fresh concrete and with steel (in the presence of chloride ions). Such reactions are harmful to concrete integrity.

8. Expansion anchors shall not be used where structural tension or moment loads are to be resisted.

Rationale: Anchorage of expansion anchors is achieved by expansion in a predrilled hole. The quality of installation is not reliable and therefore not to be used where structural tension occurs. Moreover, at higher temperatures the behaviour is less predictable.

9. Chemical anchor bolts that are installed into hardened concrete with an engineered proprietary system shall be subject to the approval of the Principal.

Rationale: Approval ensures compliance with safety standards and engineering requirements. It confirms that the chemical anchor system and the bonding material is suitable for the specific application.

10. Chemical anchor bolts used for sustained structural loads shall not be installed in overhead applications.

Rationale: The main issue is the quality of installation resulting from improper filling.

11. Epoxy anchor bolts shall not be used for surfaces that can be exposed to temperatures above 60 °C (140 °F).

Rationale: At temperature above 60 °C (140 °F) (e.g., fire or extremely hot environments) the mechanical properties of the anchorage will change, and the epoxy material might lose strength and become pliable.

2.4 LIQUID TIGHT STRUCTURES

1. Crack widths shall be limited to < 0.1 mm (0.004 in) for liquid tight structures.

Rationale: In the case of liquid tight structures, the maximum crack width is limited to 0.1 mm (0.004 in) to achieve water tightness.

2.5 PRE-STRESSED CONCRETE STRUCTURES

1. Pre-stressed concrete shall only be allowed for piles, bridges/culvert decks in roads and concrete Liquefied Natural Gas (LNG) tanks.

Rationale: The main reason to avoid pre-stressed concrete is the effect of fire. Due to high temperatures developed during a hydrocarbon fire the prestress losses will impact the structural integrity of pre-stressed structures.

2.6 FOUNDATIONS AND STRUCTURES SUBJECT TO VIBRATION

1. Foundations and structures subject to vibration shall be isolated from adjacent structures to prevent transmission of machinery vibrations.
2. Foundation analysis shall be performed in accordance with DEP 34.00.01.30-Gen.

Rationale: DEP 34.00.01.30-Gen. is the governing Shell standard for structural design and engineering of onshore facilities.

2.7 FOUNDATIONS AND STRUCTURES SUBJECT TO TEMPERATURE EFFECTS

1. Foundations and structures subject to temperature effects shall be designed for any temperature difference that could occur in parts of structural members or the whole structure.

Rationale: Ignoring temperature effects can induce stresses leading to cracking in the concrete and subsequent durability impact.

2.8 SKIRTED PROCESS COLUMN PEDESTALS AND TABLETOP STRUCTURES

1. The pedestals of the foundation for skirted process vessels shall be provided with a slope and drain.

- a. Refer to Standard Drawing S 19.014 for typical drainage details.

Rationale: In case of hydrocarbon leaks, this system works as a leak detector in the skirted area. Moreover, it prevents development of a large pool fire.

2. All edges of tabletop slabs shall have a curb of minimum 100 mm (4 in) height to enable draining the liquid to a designated drain.

Rationale: To prevent liquid and rainwater from being drained over the edge.

2.9 CHEMICAL RESISTANT PROTECTIVE SYSTEMS**2.9.1 General**

1. The Contractor shall submit a coating specification for approval by Principal that addresses the following factors:
 - a. Mechanical – load bearing functions (traffic duty definition, abrasion resistance and anti-slip properties);
 - b. Chemical – continuous and accidental chemical exposure;
 - c. Electrical – isolating, insulating, anti-static, earthing properties;
 - d. Environmental – proofing layer to enhance concrete protection under temperature, humidity or local ambient chemicals.

Rationale: Clear functional requirements are necessary to establish coating needs

2. The Contractor's coating specification shall define coating performance criteria for the following:
 - a. Service – lifetime and refurbishment activities
 - b. Constructability – crack bridging, curing methods & timing
 - c. Ambient – temperature, humidity, dust, local chemicals (nitrates, chlorides, sulphates)

Rationale: Coating should perform within design requirements and be constructible

3. Combined systems with based on synthetic-resin mortars shall not be selected where the surface is exposed to alkali or alkali solution in service.

Rationale: Presence of steam or condensate will accelerate the degradation of synthetic-resin-based mortar in alkali or alkaline solutions.

4. A protective system shall be selected and provided where there is continuous or routine direct steam impingement on concrete surfaces.

Rationale: Steam impingement into concrete accelerates degradation limiting design life.

5. Contractor shall provide the following for the selected coating system:
 - a. Material selection report, required design documents including performance criteria with laboratory reports in accordance with approved project/asset standards;
 - b. General and detailed drawings supporting construction execution for layout, layup, required termination, interface surfaces and joints in the application;
 - c. Quality control plan and execution procedures in accordance with Manufacturer/Supplier specifications;
 - d. Warranty and service life certification, required maintenance and upkeep regime as per Manufacturer/Supplier guidance.

Rationale: Execution of coating systems has to be controlled for good quality outcome with a clear view on maintenance regimes to upkeep performance

2.9.2 Design requirements

1. The Contractor shall provide all drawing details and other documentation that demonstrates how the design addresses the following:

Details and procedural drawings can be provided by Manufacturers/Suppliers.

- a. The difference in thermal expansion between the protection system and the concrete base;
- b. Meeting the Manufacturer's/Supplier's requirements for the design thicknesses of the various components;
- c. The required slope and drainage of the surface;
- d. Differential movement and load induced deflection of the concrete components;
- e. All joint designs and details are in accordance with Manufacturer's/Supplier's requirements;

Typical details include construction or contraction joints, lining termination, grating interface, pipe and cable penetrations, and drains.

- f. Changes in surface orientation (vertical to horizontal), penetrations and other interface areas where other materials bear on or are in contact with the concrete/protection system surface.

Corners, edges and contact areas historically have been where most damage to protection systems begin. These areas hold moisture longer or provide a challenge to the ability of the Applicator.

2. Isolation and expansion joints should be located outside the zones of chemical attack or where there is the least possible chance of aggressive liquids permeating them.

Rationale: To prevent damage due to chemical attack.

3. Expansion joints should not be located at the lowest point of the drainage slope.

Rationale: Joints at this location will be more vulnerable to attack from the drained liquids that accumulate there.

4. The design of the expansion joint in the reinforced concrete structure shall include both of the following:

- a. A suitable backer material: Non-absorbent to prevent moisture retention and compressible but resilient, so it accommodates joint movement;
- b. A joint filler compatible with the expected chemical exposure and temperature conditions.

Rationale: To ensure adequate durability of the expansion joint.

3. MATERIALS AND COMPONENTS

3.1 ADMIXTURES

1. Only one Manufacturer/Supplier of each admixture shall be used throughout the course of the contract scope.

Rationale: To ensure consistency in the quality and performance of the admixtures.

2. Admixtures shall be subject to approval by the Principal.

Rationale: Admixtures need to conform to chemical and performance requirements. Approval ensures quality and adherence to standards.

3. Calcium chloride admixtures and admixtures containing soluble chlorides shall not be permitted.

Rationale: Chlorides might lead to corrosion of reinforcement.

3.2 AGGREGATES

1. Aggregate shall be obtained from a single source throughout the course of the project.

Rationale: Using aggregates from a single source ensures uniformity in material properties, such as particle size distribution, gradation, and quality. This consistency contributes to predictable concrete performance. When aggregates come from a single Manufacturer/Supplier, it's easier to monitor and control their quality. Regular testing and inspection can be conducted more effectively, reducing the risk of substandard materials.

2. Aggregate shall not be gap-graded.

Rationale: The voids between big size aggregates are not properly filled by medium and small size aggregates. Hence, using gap-graded aggregate can lead to segregation and can even alter the anticipated workability. It might also impact the strength and impermeability of concrete.

3.3 CEMENT

1. Only one brand of cement shall be used throughout the course of the Project.

Rationale: To avoid mistakes at site, ensure uniformity and consistency with trial mix.

2. The cement shall be Blended Portland Blast Furnace Cement with minimum 65 % blast furnace slag, with the following exception:

- a. Blended cement with fly ash may be used provided the Contractor demonstrates that the obtained concrete mix achieves the same durability, impermeability and resistance to prevailing environmental conditions as the concrete mix based on Blast Furnace Cement with minimum 65 % blast furnace slag.

Rationale: Using Blended Portland Blast Furnace Cement results in a very dense concrete with improved durability and impermeability and reduced CO₂ footprint.

Rationale, Item 2a: Blast Furnace Slag is not available across the world.

3. If Class C fly ash is used in an aggressive sulphate environment, the mix design shall be tested to verify sulphate resistance.

Aggressive sulphate environments are defined in ACI 318 as sulfate exposure classes (S0–S3) and [EN 1992](#) (Eurocode 2) as XA aggressive chemical classes (XA1–XA3).

Rationale: Class C fly ash, which contains high levels of amorphous calcium aluminate, can make more calcium hydroxide available for reaction with sulphates, thus contributing to concrete deterioration.

3.4 JOINT SEALANT

1. Sealer shall be a two-component polysulfide compound that is resistant to products that could be spilled on the surface.

Rationale: Two-component Polysulfide sealer exhibit good resistance to various chemicals, including water, oils, grease, mild acids, and alkalis. These sealers maintain their flexibility even with building component expansion and contraction. They create a tough, rubber-like seal that can withstand stress and movement.

2. Sealer for down-hand application shall be liquid self-levelling grade.

Rationale: Liquid self-leveling sealers help achieve a smooth and level surface in down-hand applications.

3. Sealer for vertical joints or for overhead application shall be trowel-grade.

Rationale: The use of trowel-grade sealer simplifies the repair process, especially for shallow surface repairs in areas with limited access.

3.5 WATER USED DURING CONCRETING

1. The water used for mixing water, ice, curing, or any other function relating to the placement of concrete shall not contain matter which could be harmful to the reinforcement or the durability of the concrete and to the hydration of cementitious material in the concrete mix.

Rationale: To mitigate the risk of concrete degradation due to harmful substances in the water, such as chlorides, sulphates, and organic materials, that can affect the long-term durability of the concrete.

3.6 REINFORCEMENT STEEL

1. Reinforcement for general concrete work shall be hot rolled, high yield deformed bars (≥ 415 MPa).

Rationale: During hot rolling, the bars undergo deformation, resulting in surface ribs. These ribs enhance the bond between the steel and concrete.

2. Reinforcement bars shall be un-coated deformed billet steel.

Rationale: Using uncoated deformed billet steel for reinforcement bars ensures strength and effective bonding with concrete.

3. Welded wire fabric shall be flat (unrolled) plain wire fabric.

Rationale: Rolling can damage the welded wire fabric. Moreover, from a space management perspective, rolling increases the volume of transportation.

4. Tie wire use to tie reinforcement bars shall be black annealed wire with a diameter of 1.5 mm (0.06 in).

Rationale: Black annealed wire with a gauge of 1.5 mm (0.06 in) is commonly used for construction and rebar binding. Its flexibility and strength make it suitable for securing materials like reinforcement bars (rebar) in concrete work.

5. In areas where reinforcement steel congestion is anticipated, using manufactured embedment shall be subject to the approval by the Principal.

Rationale: Avoiding excessive reinforcement congestion is crucial for constructability. The design needs to allow for proper assembly and placement of all reinforcement, as well as adequate concrete casting and consolidation. In case concrete dimensions cannot be increased to avoid reinforcement congestion, mechanical anchorage can be used to ensure adequate embedment of reinforcement in concrete.

3.7 CHEMICAL-RESISTANT COMBINED (BRICK OR TILE) LINING SYSTEMS

3.7.1 General

1. The design shall address the difference in thermal expansion between the brick or tile lining, sealing layer and the concrete surface.

Rationale: To prevent damage to the joints in the chemical resistant lining.

2. The bedding mortar layer in combined systems shall be a minimum of 3 mm (1/16 in).

Rationale: To ensure required durability and to compensate for irregularities

3. The design detail for penetrations (e.g., bolts in a pump foundation or pipe support base) shall ensure the following:

- a. No aggressive media can penetrate along the bolts to behind the sealing layer;
- b. Sharp edges of the pump-base, pipe support base or ceramic materials do not damage the sealing layer;
- c. On the top of the foundation, the trowelling compound has a liquid-tight joint with the grouting of the pump.

Rationale: To prevent damage to the joints in the chemical resistant lining.

3.7.2 Sealing layers

1. Sealing layers shall provide the chemical resistance to the chemical exposure.

Rationale: To prevent damage to the joints in the chemical resistant lining.

2. Sealing layers shall be continuous and vapour-tight.

Rationale: To prevent damage to the joints in the chemical resistant lining.

3. Sealing layers shall remain flexible to accommodate expansion and contraction resulting from structural movements.

Rationale: To prevent damage to the joints in the chemical resistant lining.

4. EXECUTION OF CONCRETE STRUCTURES

4.1 EXECUTION DOCUMENTS

1. The Contractor shall propose and submit the general notes, typical details, and Project Specification for concrete works for the review and approval by the Principal.

Rationale: These submissions are essential for ensuring compliance with project requirements and standards.

2. Work shall not commence until the concrete design drawings have been approved by the Principal.

Rationale: Approval process ensures that the project proceeds smoothly, adheres to standards, and minimizes risks.

4.2 CONCRETE MIX DESIGN

1. The concrete mix design shall be submitted to the Principal for review and approval.

Rationale: Approval ensures that the mix will perform well under expected conditions (e.g., load-bearing capacity, resistance to environmental factors). It also ensures that the concrete mix meets specified strength, durability, and workability requirements.

2. The minimum cube compressive strength of the concrete mix after 28 days shall be as indicated in Table 1.

Rationale: 28 days strength benchmark is imposed by national/international codes and standards to be used in design calculations. Also, using a longer duration strength most likely will result in delays of construction.

Table 1 Minimum compressive cube strength after 28 days of curing

Item	Design Compressive Strength of Concrete
Slabs, area paving, roads, cover slabs for cable trenches, watertight concrete, machinery foundations, and concrete exposed to severe freezing and thawing conditions	28 MPa (4000 psi)
Structures and their foundations and for stationary equipment foundations except jetties	28 MPa (4000 psi)
Seal Slab or Blinding concrete, and miscellaneous non-structural concrete fill	14 MPa (2000 psi)
Concrete envelopes for underground electrical ducts	14 MPa (2000 psi)
Pre-stressed concrete structures and reinforcement concrete of nearshore and coastal structures	41 MPa (6000 psi)

3. The maximum Water Cement Ratio (WCR) shall be as specified in Table 2.

Rationale: The maximum WCR is set to ensure the required durability will be achieved.

Table 2 Maximum Water Cement Ratio

Structure	WCR maximum
Concrete foundations and structures (except as Note 1)	0.45
Seal Slab or Blinding concrete and miscellaneous non-structural concrete fill.	0.55
liquid-tight (hydraulic) structures and nearshore and coastal structures	0.40
Concrete deposited by tremie under water or for pre-stressed concrete structures	0.44
Note 1: For mixtures containing pozzolan, the WCR is the ratio of water to (cement + pozzolan).	

4.3 FORMWORK

1. Chamfer strips of 20 mm (0.8 in) shall be provided at all corners on permanently exposed surfaces.

Rationale: Chamfer strips create a 45-degree beveled edge on the concrete corners. This prevents chipping and damage during formwork removal, construction and maintenance activities.

2. Forms and shores provided for self-supporting members such as beams, floors, and walls shall not be removed until:
 - a. Cured concrete has gained at least 80 % of the specified 28-day strength;
 - b. The member is able to carry its own weight and any approved superimposed load.

Rationale: This precaution ensures the stability and safety of the structure during the construction process.

3. The Contractor shall provide a formwork striking plan that is based on the climatic conditions, heat of hydration, strength development, load conditions and relevant curing methods.

Rationale: A formwork striking plan outlines when and how formwork will be removed after casting. This plan is essential for achieving strong and durable concrete structures.

4. Contractor shall produce design calculations ensuring safety in design.

Rationale: Proper formwork design ensures safety during construction and contributes to the overall quality of the concrete structure.

4.4 REINFORCEMENT FABRICATION

1. Welding of reinforcement steel shall not be permitted.

Rationale: The heat generated during welding can compromise the strength and stability of the reinforcement, resulting in a compromised structure. Moreover, the quality of weld is questionable and difficult to check.

2. Reinforcement steel shall be bent in cold conditions (i.e., without heating the steel).

Rationale: Hot bending could soften the steel due to changes in the crystalline structure of the metal resulting in reduced yield strength.

3. Reinforcement bars which have been bent and then straightened shall not be incorporated in the work.

Rationale: Using reinforcement bars that have undergone cold bending and straightening can jeopardize the safety and durability of the structure. This is due to loss of ductility after yielding of reinforcement bars.

4.5 PLACEMENT

4.5.1 Placement of reinforcement

1. Exposed reinforcement bars, including dowels, that are vertical shall be protected using plastic protection caps.

Rationale: For safety reasons.

2. All reinforcement shall be in place before pouring concrete (i.e., no wet placement of reinforcement during concreting operations).

Rationale: All reinforcement needs to be firmly fixed in place to prevent it moving during concreting operations. Wet placement can affect concrete quality. Wet placement doesn't control the location and position of reinforcement and could result in poor embedment around reinforcement, impacting the anchorage length.

3. The minimum concrete cover to reinforcement for concrete exposed to hydrocarbons and prone to chemical attacks shall be 50 mm (2 in).

Rationale: Insufficient concrete cover for reinforcement can lead to reduced durability in the concrete structure due to chlorides penetration and carbonation.

4. To obtain the required concrete cover, concrete spacer blocks with identical compressive strength with concrete shall be used.

Rationale: Spacers need to match the compressive strength of the surrounding concrete to achieve the overall design compressive strength.

4.5.2 Placement of anchor bolts and other embedment

1. Anchor bolts, anchors, inserts, anchor rails, sleeves, drains, curb and seat angles, nosing, and other embedded items shall be positioned and securely anchored before placing concrete.

- a. Welding of these items to the reinforcement bars shall not be permitted.

Rationale: This is to prevent them moving during concreting operation and to avoid rework caused by unacceptable deviations.

Rationale, Item 1a: The heat generated during welding can compromise the strength and stability of the reinforcement, resulting in a compromised structure.

2. Anchor bolt sleeves shall be capped or filled with non-bonding mouldable material.

Rationale: To keep out water, concrete, and debris during the placement of concrete and grout.

3. Wet dipping (placement after concrete pouring) of anchor bolts shall not be permitted.

Rationale: Wet dipping is not allowed due to potential issues with improper bonding of concrete around the anchor, clashing with reinforcement and inadequate embedment length.

4. Anchor bolts shall not be heated to facilitate equipment installation.

Rationale: Heating anchor bolts can weaken their structural integrity. Heating can alter material properties of anchor bolts, affecting their load-bearing capacity, ductility and performance.

4.5.3 Concrete placement

1. Concrete temperature as delivered shall not exceed 32 °C (90 °F).
Rationale: To ensure optimal flow, curing and setting. High initial concrete temperature can lead to segregation of concrete.
2. Mixing of concrete shall not be done in concrete trucks.
Rationale: Concrete mixing needs to be done at the batching plant. Mixing in the concrete truck will jeopardize the quality of concrete and impact its strength and durability.
3. Water shall not be added to increase slump.
Rationale: Increasing the w/c ratio will decrease the concrete strength and durability performance and it will increase drying shrinkage.
4. Concrete shall not be hand-mixed or re-tempered.
Rationale: Hand mixing and re-tempering are not reliable in achieving the specified concrete strength and durability performance.
5. No additives or ingredients shall be added to the concrete after batching.
Rationale: This is to maintain consistency during concrete mixing for achieving the desired strength and durability in the final product. It mitigates the risk of modification of hardened concrete properties and uncontrolled concrete mix behaviour and quality during transporting and placing.
6. Any proposed change in mix design during the concrete work shall be subject to approval by the Principal.
Rationale: The approval process ensures that the concrete mix formulation meets specification requirements.
7. Concrete that has achieved initial set or has been contaminated by foreign matter shall not be deposited in the structure or foundation.
Rationale: Avoiding the use of partially set or contaminated concrete ensures structural safety, durability, and overall performance.
8. Concrete shall not be placed during rain, sleet, or snow unless protection is provided to avoid any changes in concrete design and material properties.
Rationale: Exposure to rain, sleet, or snow during placement can alter the water-cement ratio, affecting the concrete's properties. Excess water can dilute the cement paste, leading to reduced strength and increased permeability.
9. For underwater structures, prior to the pour, a concrete placement plan shall be submitted to Principal for review and approval.
Concrete trials to demonstrate the required quality and consistency of the concrete mix are required for underwater concrete pours.
Rationale: This plan outlines the specific methods and procedures to ensure the successful placement and curing of concrete that then ensures integrity of underwater structures.
10. A contingency plan (including facilities) to ensure the continuity of concrete placement in case the pump fails, shall be submitted for Principal review and approval.
Rationale: An approved contingency plan will include proactive measures that can mitigate the impact of pump failures and subsequently introducing of additional construction joints which will impact the structural integrity and durability performance.
11. All slump and cube test samples shall be taken from the end of the discharge line.
Rationale: Client focus is on concrete properties at the point of placement (e.g., temperature, air content).

4.5.4 Placement of pre-fabricated concrete

1. The load-bearing layer material shall be non-shrink grout.

Rationale: Non-shrink grout is stronger than concrete.

2. The load-bearing material shall be resistant to hydrocarbons or other chemicals that it could reasonably be exposed to when in service.

Rationale: This is to prevent degradation over time.

3. When pre-fabricated elements are placed in in-situ concrete slots, non-shrink grout shall be used to provide a bearing surface in the slot in accordance with the grout Manufacturer's/Supplier's specifications.

Rationale: Non-shrink grout prevents volume reduction during curing, ensuring a durable connection between the pre-fabricated element and the surrounding concrete. The strength of non-shrink grout is generally higher than strength of concrete ensuring adequate transfer of internal forces.

4.5.5 Placement of waterstops

1. Waterstops shall be installed and located according to the Manufacturer's/Supplier's recommendations, specifications and drawings as part of the formwork installation.

Rationale: Waterstops play a crucial role in preventing water infiltration through construction joints in concrete structures. To ensure their effectiveness, it is essential to follow the Manufacturer's/Supplier's guidelines, specifications and drawings during installation.

4.6 JOINTS

1. All joints shall be located and constructed in accordance with the project drawings, specifications and Manufacturer/Supplier instructions and Technical Data sheet.

Rationale: Proper placement of expansion, contraction and construction joints, in line with drawings and specifications, helps maintain the integrity of the structure.

2. Construction joints shall be perpendicular to the main reinforcement and be made and located so as to least impair the strength of the structure.

- a. Provision shall be made for transfer of shear and other forces through the construction joint.

Rationale: Generally, the construction joints are installed at the locations where bending moment is minimum.

Rationale, Item 2a: Proper construction joint design ensures structural integrity by addressing shear transfer and maintaining continuity.

4.7 TRIAL MIXES AND TRIAL POURS

1. Trial mixes and trial pours shall be conducted before the permanent placing of concrete is started to prove the following can be achieved:
 - a. Required durability/exposure condition;
 - b. Desired workability of the design mix concrete;
 - c. That the specified concrete grade can be obtained.

Rationale: A trial mix validates the design mix, considers small variations of concrete constituents to ensure the design mix is stable and meets the predicted criteria.

2. For each trial mix, inspection and testing shall be performed in accordance with the Inspection and Test Plan (ITP) developed for the project.

Rationale: Ensuring compliance with the ITP helps maintain quality and consistency in concrete production.

3. The trial mixes and pours with associated testing shall be included in the construction schedule and be completed at least one month prior to commencement of concreting works.

Rationale: Timely execution prior to commencing of concrete works will avoid any unnecessary delay in construction schedule.

4.8 MASS CONCRETING

1. For mass concreting, the maximum peak temperature in the inside of the concrete pour shall not exceed 70 °C (158 °F).

Rationale: This limit is crucial for preventing delayed ettringite attack and thermal cracking.

2. For mass concreting, the temperature gradient between the inside and outside faces of the concrete shall not be more than the following as applicable:

- a. 20 °C (36 °F) for concrete pours subject to limited external restraining;
- b. 15 °C (27 °F) for concrete members subject to high external restraining.

Rationale: Excessive temperature differentials can lead to thermal cracking. Thus, maintaining an appropriate temperature gradient is crucial for structural integrity and longevity.

3. Curing duration for mass concreting shall be as per Section 3.3 of ACI 308.

Rationale: Curing duration is essential to ensure the temperature gradient between the inside and outside faces will not exceed the specified limits.

4. A mock-up test using the following steps shall be performed to validate the mix design:

- a. Perform heat dissipation studies to confirm that the heat of hydration will not cause a peak temperature and a temperature differential between the interior of the concrete and any outside face greater than the values specified in (4.8, Item 1) and (4.8, Item 2).
- b. Monitor the heat of hydration (using thermocouples) in trials to demonstrate excessive thermal gradient is not present during the works.
- c. Monitor strength development versus time.
- d. Observe bleeding and the general condition of the concrete during the pour.
- e. Implement precautionary measures, insulation and curing to avoid crack formation.
- f. After removal of formwork, check on crack formation.

Rationale: It is essential to perform a mock-up test to develop a control method statement for the actual pouring at the construction site.

5. The mock up piece shall be of 1:1 scale (full scale) representing actual pour conditions, thickness, formwork and insulation type, formwork removal conditions, ambient radiation/temperature conditions and physical restraints.

For foundations/slabs where a 1:1 scale test sample is not possible from construction point of view, a smaller test sample can be tested if the side faces of the samples are provided with 100 mm (4 in) thick polystyrene insulation to simulate the much larger lateral dimension of the actual concrete member, keeping the sample thickness the same as the actual member.

6. The mock up test results/observations shall also be used to develop a method statement for concrete pouring at site.
7. For mass concreting mix design, the method statement shall be subject to approval of the Principal.

Rationale: This ensures that the specified procedures and practices align with the project requirements and safety standards.

8. After mass concreting, when forms are removed before the specified curing is completed, measures shall be taken to continue the curing and provide adequate thermal protection for the concrete.

Rationale: Differential temperatures can lead to development of cracks and hence adequate thermal protection is mandatory in case of early form removals.

4.9 EXTREME WEATHER PROVISIONS

4.9.1 General

1. Provisions for pouring concrete in extreme weather conditions shall be subject to approval by the Principal.

Rationale: Approval by the Principal ensures compliance with best practices.

4.9.2 Hot weather concreting

1. Concrete placement shall be in accordance with the requirements of ACI 301 and ACI 305R when the ambient temperature exceeds 32 °C (90 °F).

Rationale: When placing concrete in hot weather conditions, it's essential to follow industry guidelines to ensure proper performance and durability.

4.9.3 Cold weather concreting

1. Concrete placement shall be in accordance with the requirements of ACI 301 and ACI 306R when the ambient temperature drops below 4 °C (40 °F).

Rationale: When the ambient temperature drops below 4°C (40°F), certain precautions, as per industry guidelines, are necessary to ensure the quality and durability of the concrete. Adherence to these guidelines helps prevent issues related to concrete strength development, cracking, and durability.

2. The following measures shall be provided for curing when ambient temperature could drop below 4 °C (40 °F):
 - a. Area warming and humidification control of the curing of concrete continues until it develops more than 85 % of the compressive strength.
 - b. Concrete is allowed to dry for at least 12 hours before discontinuing or removing the temperature protection.

Rationale: When curing concrete in cold weather, it is essential to take the specific precautions to ensure its strength and durability.

3. If freezing or thawing is anticipated at any time during construction or in service, when either the cement paste or the aggregates are critically saturated near the surface of the concrete, air-entrained concrete shall be provided.

Rationale: When curing concrete in cold weather, it is essential to take the specific precautions to ensure its strength and durability.

4.10 FINISHING

1. The finish shall be without objectionable surface fins, holes or rough patches.

Rationale: Concrete finishes play a crucial role in achieving both functional and aesthetic goals, making them an integral part of construction and design. Moreover, it ensures safety during construction and operations, e.g., slippery.

2. The surface of the concrete shall be finished as indicated in Table 3.

Table 3 gives required finish based on use and location.

Table 3 Concrete surface finish

Surface	Finish
Surface requiring plastering or tiling	Scratched
Area Paving	Broom or Belt
Compressor house floors	Broom or Belt
Cooling tower Basin, Flume	Floated
Curbs	Broom or Belt
Foundations and pedestal to be grouted	Scratched
Equipment bases	Floated
Platforms, Stair Pads, Concrete Stairs	Broom or Belt
Floor Slabs	Trowel
Formed Surfaces	As-Cast Finish
Suction Pits	Floated

4.11 LIQUID TIGHT STRUCTURES

1. Liquid tight structures shall be cured by a wet-cure procedure only (e.g., use of ponding, sprinkling, or a moisture-retaining fabric).

Rationale: Curing liquid tight structures using a wet-cure procedure is essential for their long-term performance and durability.

2. Moist wood forms in contact with concrete shall not be considered as a curing method for liquid tight structures.

Rationale: It is not a controlled method of curing and therefore the quality is questionable.

3. Immediately after placement, concrete for liquid tight structures shall be protected from premature drying, excessive high or low temperatures, rain, flowing water or mechanical damage.

Rationale: After concrete is freshly placed, it is crucial to protect it from these factors to ensure its quality and durability.

4. Curing shall occur for a minimum of 14 days for liquid tight structures.

- a. Wall forms shall then be loosened, and water continually sprinkled between the wall and forms.

Rationale: During the initial 14 days, concrete gains approximately 90% of its total compressive strength. Hence, proper curing for initial 14 days is critical for strength gain and tensile strength.

Rationale, Item 4a: This ensures process of maintaining satisfactory temperature and moisture conditions in concrete long enough for hydration to develop the desired concrete properties.

5. The surface of concrete at all joints shall be thoroughly cleaned with all laitance removed before placing adjoining concrete.

Rationale: This requirement ensures good bonding and structural integrity. Moreover, water-tightness test will accelerate self-healing process of small cracks.

6. Chemically resistant coatings in accordance with (4.16) and (5) shall be used to protect concrete surface from degradation by harmful liquids.

Harmful liquids to concrete include salt solutions (e.g., aluminum chloride, ammonium nitrate, ammonium sulfate, sodium sulfate, magnesium sulfate, calcium sulfate), alkaline solutions (e.g., 20 wt % or greater solutions of sodium hydroxide or potassium hydroxide) and all organic and inorganic acids other than oxalic and tartaric acids.

7. Liquid tight structures shall pass a water-tightness test, according to a method statement as identified in the ITP.

Rationale: Liquid tight structures, such as reservoirs, basins, and tanks, need to pass a water-tightness test to ensure they can effectively hold liquids without leakage.

4.12 PRECAST CONCRETE

1. Precast members shall not be lifted or handled within 72 hours of the completion of pouring and finishing.

Rationale: This requirement ensures the safety of workers and prevent accidents during the handling and installation of precast concrete elements. Moreover, this requirement ensures a minimum strength is achieved required to absorb safely the stresses caused by lifting activities.

2. Structures and members shall be handled, lifted and supported so that it does not compromise the integrity of the member.

Rationale: Proper handling eliminates accidental overstressing.

3. A lifting and erection procedure for precast facilities shall be subject to review and approval by the Principal.

- a. Once accepted, no other method shall be used during the whole period of construction, unless approved by the Principal.

Rationale: When dealing with precast facilities, it is essential to follow proper lifting and erection procedures to ensure safety, structural integrity and quality.

4. Connection system between precast structural members shall be subject to approval by the Principal.

Rationale: Principal approval ensures that the connection system meets safety, design, and performance criteria for precast structures.

5. Adequate bracing or propping shall be provided for precast structural members until permanent connections are made.

Rationale: To prevent instability or collapse of the precast members during construction, as they cannot safely support loads until permanent connections are secured

4.13 USE OF SELF-COMPACTING CONCRETE (SCC)

1. The mix design for SCC shall ensure a dense, homogeneous concrete with at least the same engineering properties of durability, impermeability and resistance to prevailing environmental conditions as traditional vibrated concrete.

Rationale: The design mix of self-compacting concrete aims to achieve durability against chemical and environmental aggressiveness.

2. Due to the water reducing effect of the special admixtures used in the SCC mix, the sensitivity for moisture variation in the aggregates shall be addressed in the trial mixes.

Rationale: Addressing aggregate moisture variation during trial mixes ensures that the concrete mix performs as expected, balancing strength, workability, and durability. Moreover, it ensures a stable mix which will behave adequately under moisture variation.

3. Because of the high workability of SCC mixes, the formwork shall be watertight and capable of withstanding the pressure against it during casting.

Rationale: SCC's high flowability allows it to flow easily during casting, making it essential to have water-tight formwork capable of withstanding the pressure exerted by the fluid concrete and to avoid honeycombing.

4.14 FIBRE REINFORCEMENT CONCRETE

1. Fibre reinforcement concrete shall only be used for non-structural items such as paving.

Rationale: For structural elements, conventional reinforcement (such as steel bars) is preferred because it offers higher load-bearing capacity, constructability and efficient execution. While fibre reinforcement concrete can enhance non-structural elements like paving, structural applications often require additional reinforcement.

2. The reinforcement fibres shall be evenly dispersed throughout the concrete during the mixing process.

Rationale: Evenly distribution is important to ensure tensile strength requirement is achieved.

4.15 REPAIR

1. Before starting any concrete work, a repair method statement shall be developed and submitted to the Principal for review and approval.

Rationale: A repair method statement is essential for ensuring the safe and effective restoration of damaged concrete structures.

2. The repair materials shall be based on cement and not epoxy resin to ensure compatibility with parent concrete.

Rationale: Due to different material parameters, (e.g., much higher coefficient of thermal expansion and much lower E-modulus) epoxy material in many cases is unsuitable as repair material for concrete facilities.

3. Concrete surfaces shall not be repaired or touched up after the removal of formwork until the surfaces have been inspected and released in accordance with the quality procedures.

Rationale: This ensures that any necessary repairs or finishing operations are performed correctly and that the concrete remains structurally sound.

4.16 CHEMICAL RESISTANT PROTECTION SYSTEMS

1. All materials shall be handled and stored in accordance with the Manufacturer's/Supplier's requirements.

Rationale: To maximise the useful life of the materials.

2. All materials shall be protected against weathering conditions until being placed, including exposure to direct sunlight or frost.
 - a. In climates where it is impractical to provide continuous protection against freezing, materials should be defrosted and warmed in accordance with the Manufacturer/Supplier recommendations.

Rationale: To maximise the useful life of the materials.

3. The shelf life of materials shall be documented and monitored in accordance with the Manufacturer/Supplier requirements.

- a. Materials shall be stored and handled on a basis of first-in/first-out.

Rationale: To maximise the useful life of the materials.

4. In the absence of clear guidance from the Manufacturer/ Supplier, materials that have been stored for a period longer than six months shall be subjected to new quality control tests.

- a. A test report shall be submitted to the Principal before using these materials.

Rationale: To ensure that the materials still can be used without any degradation of performance.

5. Combined system bricks and tiles shall be handled, unloaded and stacked by hand or by using a pair of brick tongs.

Rationale: To ensure that the bricks and tiles are not damaged during handling, unloading and stacking.

6. Cements of different brands shall not be mixed.

- a. Labels indicating the name, quality and quantity of the contents of cement containers shall not be removed until final use.

Rationale: Mixing different cements will lead to spatial inconsistencies in strength and settling time.

7. The Contractor shall define the controls for separation and segregation of ingredients or components of the protective system from others in a material handling document that is shared with all Applicators.

Rationale: Some components from one type material (e.g., grout constituents) might cause contamination and render a batch unusable.

5. INSTALLATION OF CHEMICAL RESISTANT PROTECTION SYSTEMS

5.1 GENERAL

1. The Contractor shall prepare documentation (including schematics) that addresses the following:
 - a. preparation for and execution of each step of the installation;
 - b. the required competence of the Applicator to ensure quality and performance of the concrete protection system.

Rationale: To align on how the chemical resistant protection system will be installed.

2. The documented installation steps and competence requirements shall be in accordance with Manufacturer's/Supplier's requirements.

Rationale: To ensure proper installation of the chemical resistant protection system.

3. The required timing of each preparation and installation step relative to its predecessor shall be documented.

Rationale: To prevent damage because an installation step was done before the curing process is completed.

4. Until protective systems are fully cured and handed over to the Principal, they shall be protected against mechanical abuse, welding activities, scaffolding and detrimental weather effects.

In many cases it is advantageous to leave this protection until full commissioning of the plant area.

5.2 SURFACE PREPARATION FOR CONCRETE SURFACES

1. The requirements for application of the protection system on new concrete shall define acceptance criteria for the following conditions of the concrete surface in accordance with Manufacturer's/Supplier's requirements:

- a. moisture;

Typically, moisture content from the surface (up to approximately 20 mm (3/4 in) deep) is kept less than 4 % by volume.

- b. cracks;
- c. joints in concrete (e.g., construction, contraction, expansion, and isolation);
- d. surface orientation inclined to vertical surfaces;
- e. surface flatness;
- f. slope;
- g. composition, strength and related cure status;
- h. surface dust, laminate, oil, grease or other contaminant;
- i. temperature;
- j. size and depth of surface defects (e.g., fins, air holes, honeycombs);
- k. surface roughness;
- l. pH.

Rationale: To ensure adequate quality assurance and quality control.

2. In liquid retaining structures, cracks wider than 0.10 mm (4 mil) shall be repaired by synthetic resin injection.

Rationale: To prevent damage due to the cracks.

3. The final surfaces shall be smooth and even without any sharp edges, burrs, (radius > 5 mm (0.2 in)) in compliance with the preparation requirements specified by the Manufacturer/Supplier.
4. The Contractor shall prepare and submit to the Principal the repair techniques needed to mitigate possible surface characteristics that are not per initial acceptance criteria.

Rationale: To align on the actions needed to remedy the defects.

5. Concrete which has already been damaged by chemical products shall be neutralised with all damaged material removed.
 - a. Rehabilitation measures shall be developed in consultation with the Principal.
6. For rehabilitation of concrete, the Contractor shall develop a step by step method to prepare the existing concrete surfaces to the acceptance criteria.
7. If concrete is coated with a phenol-free bitumen coating, the dry film thickness of the coating shall be at least 400 µm (16 mil).

Rationale: To ensure required durability, structural integrity and proper levelling

5.3 COMBINED SYSTEMS

5.3.1 Sealing Layers

5.3.1.1 General

1. The sealing layers shall be clean and free of dust, oil, grease or other contaminants before subsequent application layers.

Rationale: To prevent inadequate coverage of the subsequent layers due to the presence of contaminants.

2. The concrete surface shall be primed in accordance with the requirements of the Manufacturer/ Supplier before application of the sealing layer.

Rationale: To ensure proper application.

5.3.1.2 Modified bitumen sealing layers

1. The modified bitumen shall be spread by squeegees or brush until it is smooth and even.

Rationale: To ensure adequate quality of layer thickness.

2. The surface of the modified bitumen sealing layer shall be sanded for good adhesion of the subsequent mortar layer.

5.3.1.3 Thermosetting sealing layers

1. For proper adhesion of the thermosetting sealing layer, the concrete substrate shall be roughened before applying a primer.

Grit blasting, pressurized water blast-cleaning or shot peening can be used to roughen the surface.

Typically within 24 hours after surface preparation, the epoxy resin and the glass fibre reinforcement (if any) is applied to the specified thickness.

Rationale: To ensure the primer adequately adheres to the substrate.

5.3.2 Mortars

5.3.2.1 General

1. Prior to the application of mortars directly to a concrete or brickwork substrate, the surfaces shall be made liquid-tight with a primer.
 - a. The primer shall be in accordance with the requirements and recommendations of the Manufacturer/Supplier.

Additional sanding might be required to improve adhesion of subsequent mortar layers.

Rationale: To prevent damage due to liquids penetrating into the surface.

2. If the mortar type is supplied in two components, they shall be thoroughly mixed and used immediately.

For example: silicate cement, phenol-furfuraldehyde resin, polyester resin.

To avoid using mortar that has already started to cure, only a limited quantity is typically mixed at a time.

Rationale: To ensure the mortar is applied before it sets and cures.

3. The installation method shall include management of the storage and mixing of components according to ambient conditions.

Rationale: To prevent contamination or degradation of the components.

5.3.2.2 Mortars based on hydraulic cements

1. Layers of mortars based on hydraulic cement, applied to provide slopes, shall be kept wet during curing to obtain optimum strength and to avoid hairline cracks.

A wet state during curing is typically for about one week.

2. Hydraulic cement supplied in paper bags shall be used within 8 hours of opening the bag.

Rationale: To avoid starting the hydration reaction with moisture in the air.

3. The mortar used as bedding mortar shall have a cement/sand ratio of 1:3 by volume.

Rationale: A higher content of sand will impact the packing of constituents and therefore the strength whereas a higher content of cement will result in higher shrinkage causing cracks.

5.3.2.3 Mortars based on phenol-furfuraldehyde resin or furane resin

The mortar is supplied in two components, a liquid and a powder.

1. The substrate temperature shall be managed throughout the cure using electric heaters as recorded in work method statement.

- a. To begin application, substrate temperature shall not exceed 80 °C (176 °F).

Rationale: The difference in expansion between tile, substrate and the top surface will adversely affect adhesion.

2. The grout shall be protected from contact with water or water vapour during curing.

Rationale: Water or water vapour impact the strength and chemical resistance of grouts based on chemical reaction and not hydration.

3. During the curing, the mortar shall not come into contact with free alkali (e.g., if the seal layer is an alkali mortar), as this would tend to neutralise the acid catalyst.

- a. If alkali mortar is used as a sealing layer, two coats of a Manufacturer/Supplier approved primer should be used as a sealant between the materials.

Rationale: To prevent contact between the resin mortar and alkali-based mortars, which have a high pH due to cementitious materials. A primer serves as a protective barrier and applying two coats will increase its effectiveness.

4. Furane resin-based mortar shall not be applied directly to concrete surfaces without a primer approved by the Manufacturer/Supplier.

Rationale: It is common practice to use primers to ensure adequate adhesion between concrete and resin mortar. Manufacturer's/Supplier's approval is required to ensure compatibility with resin mortar

5.3.2.4 Mortars based on polyester resin

1. The components, (in the form of a powder and a liquid resin), shall be used immediately after being mixed.

The components are self-curing at 15 °C – 20 °C (59 °F – 68 °F); a complete cure at this temperature can be obtained in 24 hours. For optimum chemical resistance a longer curing period is usual. The curing time and pot life are affected by temperature.

2. The grout shall be protected from contact with water or water vapour during curing.

Rationale: Water or water vapour impact the strength and chemical resistance of grouts based on chemical reaction and not hydration.

5.3.2.5 Mortars based on epoxy resin

These mortars are generally supplied as a paste of putty-like consistency, together with a liquid curing agent.

After the two components have been mixed the mortar cures within one hour at temperatures of 10 °C – 30 °C (50 °F – 86 °F). The curing time is affected by temperature. The curing agent generally used is a cold-curing type, which limits the maximum operating temperature but facilitates processing. In tropical conditions a hot-curing type is typical due to its prolonged pot life.

1. The grout shall be protected from contact with water or water vapour during curing.

Epoxy resin-based mortars have very good chemical resistance. If filler such as graphite is added, resistance to hydrofluoric acid is also obtained.

Rationale: Water or water vapour impact the strength and chemical resistance of grouts based on chemical reaction and not hydration

5.3.3 Bricks and tiles

5.3.3.1 General

1. Bricks and tiles shall be clean and dry during application.
Rationale: To prevent damage and misalignments due to the presence of debris or other contaminants.
2. When brick or substrate temperature will be below 15 °C (59 °F), hoarding enclosure and area heating shall be provided to maintain temperatures above this limit.
Rationale: To avoid issues associated with poor adhesion, cracking, or improper workability and curing of mortar or adhesives which could occur when brick or substrate temperature is below 15 °C (59 °F).
3. The brick lining shall be protected from rain, snow, frost and hail until the system is complete and cured.
Rationale: Moisture and cold temperatures will damage or otherwise compromise the lining
4. The bricks or tiles shall fit squarely and within tolerance in compliance with Manufacturer/Supplier requirements.
5. When a sealing layer is present, the bottom shall be protected first with a brick layer.
Rationale: To prevent mechanical damage and to ensure specified performance of sealing layer.
6. When the bottom is to be placed first, the perimeter bricks of the horizontal surface (especially those under a future vertical surface) shall be installed first as a shell around the rest of the bricks to be placed within that perimeter.
Rationale: Generally, without a seal layer present the vertically placed lining is held in place by the bottom layers, which eases repair of the generally higher (mechanical) loaded floor tiles. The shell however allows the same maintenance stability.
7. In a multi-layer application, at least the top layer of the bottom lining shall be placed as described in (5.3.3.1, Item 5) and (5.3.3.1, Item 6).
8. Acid-resistant bricks or tiles shall be applied to pump foundations before the bricks or tiles are laid on the adjoining floors.
Rationale: Pump foundation allows proper alignment and bonding between the pump foundation and the surrounding floor lining, avoiding gaps that could lead to chemical penetration. Moreover, if the bricks or tiles are laid on adjoining floors first then quality issues could occur because the access to the pump foundation might be restricted.

5.3.3.2 Jointing bricks and tiles

1. Bricks and tiles shall be fully laid in mortar (bed and cross joints), except when cross joints between bricks or tiles will have a different kind of mortar, (e.g., with enhanced resistance against the envisaged chemical loads).
Rationale: Fully laid bricks in the mortar will eliminate any gaps and voids which can result in movement of the bricks.
2. Brick and tile joints shall be free of air inclusions.
Rationale: Air inclusions could get filled with aggressive chemicals, or can exert uncontrolled high pressures at elevated temperatures.
3. The bed joint between the bricks or tiles and the sealing layer shall have a minimum thickness of 6 mm (1/4 in).
Rationale: To ensure required durability and compensate irregularities.

4. The rough surfaces to be embedded shall have no edge protrusions (i.e., from split tiles or hand-cut bricks) larger than 3 mm (1/8 in), to assure that sufficient mortar thickness remains present between the bricks or tiles and the sealing layer.
5. The width of cross joints between bricks or tiles shall be between 3 mm (1/8 in) and 6 mm (1/4 in) with the following exception:
 - a. The width shall be between 6 mm and 9 mm (1/4 in and 3/8 in) for hot pour jointing materials or if the joints require sealing.
 - b. When re-jointing is required after a period of service, the joint shall be made at least 5 mm (3/16 in) wide.
 - c. The width of the cross joints shall be consistent over the full depth of the joint.

Rationale: To ensure required durability, structural integrity and compensate irregularities.

6. The joint filling shall be free of cavities and air inclusions.

Rationale: Cavities and air inclusions could get filled with aggressive chemicals, or can exert uncontrolled high pressures at elevated temperatures.

5.4 JOINTLESS SYSTEMS

5.4.1 General

The following descriptions mainly relate to application on floors but can also be applicable to adjacent walls, corbels, foundation blocks, and columns which are subject to the same chemical loads.

1. The floor coating shall be applied in accordance with the Manufacturer's/Supplier's requirements.
2. The floors shall have a slope to permit the drainage of rainwater and spillage water.
3. The floors shall be provided with a non-slip surface layer where frequent access will be required.

Surfaces can be made slip resistant by scattering sand, powdered quartz or silicon carbide on the wet surface immediately after placement.

4. Jointless systems shall be continuous, liquid-tight and resistant to the chemicals (and concentration) to which they could become exposed.

Rationale: To prevent damage due to liquid contamination.

5. To obtain good adhesion between the flooring and the concrete, the concrete shall be sealed with one or more coats of an unfilled solvent-free resin primer in accordance with the Manufacturer's/Supplier's requirements.

The unfilled liquid component of the flooring compound is typically used as a primer.

6. A self-levelling floor coating shall be applied in one layer with a thickness varying between 2 mm - 4 mm (80 mil – 160 mil).

- a. If concrete is coated with a phenol-free bitumen coating, then the dry film thickness of the coating shall be at least 400 µm (16 mil).

Rationale: To ensure required durability, structural integrity and proper levelling.

7. At least the top coat shall be UV-resistant in case of exposure to the environment.

Rationale: To prevent degradation due to UV exposure.

8. If a non-slip surface layer is required, sand or powdered quartz or carborundum shall be scattered on the material at the moment of gelling of the floor coating, so that it will not sink into the material.

5.4.2 Application of the trowelling floor coating

1. The trowelling floor coating shall be applied wet in wet until the specified thickness has been obtained.

Rationale: To avoid cold joints and to ensure a monolithic floor coating.

5.4.3 Application of the self-levelling floor coating

1. The application shall be carried out by casting or spraying to the specified thickness.

5.4.4 Application of a glass fibre reinforced synthetic resin

1. The thickness of glass or synthetic fibre-reinforced synthetic-resin based lining systems shall be at least 0.40 mm (16 mil).

Rationale: To ensure adequate embedment of glass or synthetic fiber reinforcement.

2. After preparation of the concrete surface, a primer consisting of a mixture of resin and curing agent shall be sprayed or brushed onto the concrete.

Rationale: Spraying or brushing ensures proper penetration of the primer into concrete pores and even thickness of the area whereas rolling often results in air pockets and uneven thickness.

3. Layers of glass fibre or synthetic fibre reinforcement, after being impregnated with synthetic resin, shall be applied until the specified thickness has been obtained.

- a. Glass fibres or synthetic fibres shall be chosen in accordance with the chemical load.

5.4.5 Application of a synthetic-resin-based coating system

The coating is applied to a clean and dry concrete surface by means of a brush, roller or spraying equipment.

1. The system shall consist of a primer and two or more coats of high-build coating, applied to a total dry film thickness of at least 0.40 mm (16 mil).

Rationale: To ensure required durability, chemical resistance structural integrity and thickness control.

5.5 TYPICAL APPLICATIONS**5.5.2 Expansion joints**

The typical expansion joint in the concrete is 10 mm (1/2 in) wide.

1. The expansion joint provided in the reinforced concrete shall be filled with semi-rigid polyurethane foam, insulation cord, or other appropriate material that maintains the required aspect ratio of the sealant and bend radius of the foil, while providing freedom of movement for the concrete.

Rationale: To prevent damage to the joint.

2. The joint design shall ensure that the joint remains flush with the adjacent surface of brick or tile.

- a. The joint shall be filled with a chemical or temperature resistant elastic sealing compound that is compatible with the base materials and resistant to the service exposure.

Rationale: To prevent damage to the joint.

5.5.3 Concrete trenches

Drains, sewer systems and pits are often attacked by chemical products present in the waste water.

1. The slope of a trench shall be maintained after installation of the liner or coating material.

Rationale: To ensure that the trench remains fully functional once the liner or coating material has been applied.

2. The trench shall be wide enough to ensure that the coating or lining installation can be properly installed.
3. When a trench is constructed in an acid-resistant floor, the sealing layer of both the floor and the trench shall be continuous.

Rationale: To prevent damage to the trench.

5.5.4 Pump foundations

1. If the pump foundation needs to be protected, then the sealing layer of the floor and that of the pump foundation shall be continuous, including under the pump base.

Rationale: To prevent damage to the pump foundation.

6. QUALITY PLANS**6.1 GENERAL**

1. Written quality plans and procedures shall be submitted to the Principal for review and approval.

The quality plans and procedures provide details of how compliance with the requirements of this DEP, drawings and specifications is to be achieved.

Rationale: Submitting written quality plans and procedures to the Principal is an essential step in ensuring consistent standards and practices within an organization. These plans serve as a roadmap for quality control and process improvement.

2. Quality plans shall address all aspects related to local conditions, such as climatic conditions, back up facilities, spare parts, transport possibilities, storage facilities, quarries, local Manufacturers/Suppliers, test facilities (field laboratory).

Rationale: This is to ensure that quality plans align with overall strategic goals and ensure consistent quality across products or services.

3. Materials should be obtained from the same sources throughout the work (including both trial and production mixes).

Rationale: This practice ensures consistency and reliability in the concrete manufacturing process.

6.2 INSPECTION, TESTING AND REPORTING

1. ITPs identifying the tests, frequencies, rejection criteria and responsibilities shall be prepared as part of the quality plan.

Rationale: An ITP is a crucial document in construction quality management. It outlines when, what, and how quality inspections will occur throughout a project.

2. Tests for approval and acceptance of materials used for concrete production as well as tests on reinforcement shall be performed by a certified testing laboratory approved by the Principal.

Rationale: Certified testing ensures quality and compliance.

3. A reporting system of Quality Control Records (QCRs) shall be part of the quality plan.

Rationale: A reporting system for QCR helps monitor and improve the quality of concrete construction works.

4. The reporting system shall record all results obtained during the following:

- a. Inspection of concrete members;
- b. Material and concrete testing;
- c. Material and concrete transporting;
- d. Concrete production;
- e. Concrete supplying;
- f. Concrete placing/pouring;
- g. Concrete finishing;
- h. Concrete curing;
- i. Formwork striking.

Rationale: The reporting system for concrete construction ensures quality.

5. Weather conditions shall be recorded in the quality records.

Rationale: Quality of concrete installation could be impacted by weather conditions. Hence, weather conditions are recorded in quality records.

6. The proposed quality plans shall cover the following:

- a. Setting out;
- b. Anchor bolts;
- c. Rebar and cover;
- d. Concrete strength;
- e. Heat of hydration for mass concrete;
- f. Inserts and other embedments;
- g. Formwork (including loading capacity, if required);
- h. Concrete placing;
- i. Field concrete workability;
- j. Curing;
- k. Removal of formwork (take loading capacity into account);
- l. Joint preparations- defects.

Rationale: The ITPs and QCRs are essential components of a Quality Plan for construction projects. They help ensure that the concreting work meets specified standards and requirements for the concrete construction.

7. After completion, if there is any doubt about the concrete member integrity based on the inspection records, rebound or ultrasonic measurements shall be taken to control and verify the concrete quality against the compliance tests and required concrete quality.

- a. Any deviation shall be reported, including the measurements to be taken.

Rationale: Industry wide studies have been conducted to correlate Non-Destructive Testing (NDT) results (such as Ultrasonic Pulse Velocity (UPV) and rebound hammer values) with the compressive strengths obtained from compression tests. These correlations help estimate the concrete strength based on non-destructive measurements.

Rationale, Item 7a: Prompt deviation identification, documentation, and investigation are crucial to determine their impact on product quality and safety, leading to appropriate corrective and preventive actions.

6.3 QUALITY CONTROL FOR CHEMICAL RESISTANT PROTECTION SYSTEMS

1. The application of chemical-resistant linings shall be done as per Manufacturer/Supplier specifications and procedural guidance to assure performance as per design.
2. The Contractor shall prepare a work method statement and conduct a field trial before work scope execution for specialist linings and coating applications with prior Principal review.

Rationale: To align on how the lining and coating will be applied

3. Contractor shall prepare an inspection test plan for each chemical-resistant protective system application to cover following aspects:
 - a. Manufacturer/Supplier verification;
 - b. Material transport;
 - c. Site receipt verification;
 - d. Material storage;
 - e. Surface preparation;
 - f. Application of coating system as per method statement
 - g. Certifications as required by local regulations;
 - h. Final inspection;
 - i. Service test for commissioning.

Rationale: To ensure adequate quality assurance and quality control.

4. The ITP shall address the following items during application of the sealing layer:
 - a. sufficient adhesive;
 - b. air inclusions or other visible defects;

In pits or tanks, final inspection can be carried out after hydraulic or vacuum testing.

- c. damages as result of installation accidents;
 - d. faults of the seams or of overlapping;
 - e. The sharpness in the edges of bricks and tiles (especially cut tiles).

Typically, acceptance is that edges have a radius > 3 mm (1/8 in).

5. The ITP shall address the timing of works as tied to acceptance testing.
 - a. Brick lining shall not commence until the applied sealing layer has been inspected and accepted by the Principal.

6. The final brick lining of a combined system shall be inspected for the following conditions:
 - a. General appearance of the brick lining.
 - b. The specified dimensions of the joint.
 - c. The execution of the jointing.
7. The Applicator shall have SSPC or equivalent qualification for applying the jointless system and the sealing layers of the combined protection systems.
 - a. The individual Applicator of a jointless system shall be certified or approved by the Manufacturer/Supplier confirming the individual has the required competency to apply their product.

Rationale: To ensure adequate quality assurance and quality.

8. The ITP requirements for sealing layers of combined systems shall include test criteria listed in Table 4.

Table 4 Summary of main requirements for sealing layers of combined systems

Property	Test method/ criteria	Modified bitumen sealing layer	Plastic sealing layer
Surface condition/ Visual examination	No surface defects.	x	x
Adhesion	Check by careful knocking. No lack of adhesion.	x	x
Porosity	Visual examination. No porosity.	x	x
Thickness (Note 1)	Physical measurement.	x	x
Curing	No softening after 1 minute rubbing with acetone.	No curing requirements	x (Note 2)
NOTES: 1. Depending on the type of sealing layer, thickness is measured before or after application. 2. Applies only to glass fibre reinforced epoxy-resin-based sealing layers because it directly measures the cure quality of epoxy resin systems, which is essential for their performance. Other plastic materials (e.g. PE) do not soften with solvent cure, so this test will not provide meaningful results.			

7. REFERENCES

In this DEP, reference is made to the following publications:

Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

SHELL STANDARDS

DEP 00.00.05.80-Gen.	Offline feedback form
DEP 34.00.01.30-Gen.	Structural design and engineering of onshore structures
DEP 34.28.00.31-Gen.	Onshore steel structures
DEP 35.00.10.10-Gen.	Design of jetty facilities (amendments/supplements to BS 6349-1-1/1-2/1-3/1-4, BS 6349-2 and BS 6349-4)

STANDARD DRAWINGS

S 10.047	Standard anchor bolt details
S 19.014	Octagon foundation drain details

AMERICAN STANDARDS

ACI 301	Specifications for Structural Concrete
ACI 305R	Guide to Hot Weather Concreting
ACI 306R	Guide to Cold Weather Concreting
ACI 308	Standard Specification for Curing Concrete
ACI 318	Building Code Requirements for Structural Concrete and Commentary

EUROPEAN STANDARDS

EN 1992	Eurocode 2: Design of concrete structures
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